

Exercise Sheet 10 - Solutions

Predicate Logic – Equivalences & Semantics

1. (a) We first prove the left-to-right implication:

$$\begin{array}{c}
 \frac{\frac{\overline{\forall x.(p(x) \vee \exists y.q(x, y))}}{p(z) \vee \exists y.q(z, y)} \quad 1 \quad [\forall E] \quad \frac{\overline{p(z)}}{p(z) \vee q(z, z)} \quad 2 \quad [\vee I_L] \quad \frac{\overline{p(z) \vee q(z, y)}}{\exists y.(p(z) \vee q(z, y))} \quad [\exists I] \quad \frac{\overline{q(z, y)}}{p(z) \vee q(z, y)} \quad 4 \quad [\vee I_R] \quad \frac{\overline{\exists y.q(z, y)}}{\exists y.(p(z) \vee q(z, y))} \quad 3 \quad [\exists I] \quad \frac{\overline{\exists y.(p(z) \vee q(z, y))}}{(\exists y.q(z, y)) \rightarrow \exists y.(p(z) \vee q(z, y))} \quad 4 \quad [\exists E] \\
 \frac{p(z) \rightarrow \exists y.(p(z) \vee q(z, y)) \quad 2 \quad [\rightarrow I] \quad (\exists y.q(z, y)) \rightarrow \exists y.(p(z) \vee q(z, y)) \quad 3 \quad [\rightarrow I] \quad \frac{(\exists y.q(z, y)) \rightarrow \exists y.(p(z) \vee q(z, y))}{\exists y.(p(z) \vee q(z, y))} \quad [\vee E] \\
 \frac{\exists y.(p(z) \vee q(z, y))}{\forall x.\exists y.(p(x) \vee q(x, y))} \quad [\forall I] \\
 \frac{\forall x.\exists y.(p(x) \vee q(x, y))}{(\forall x.(p(x) \vee \exists y.q(x, y))) \rightarrow (\forall x.\exists y.(p(x) \vee q(x, y)))} \quad 1 \quad [\rightarrow I]
 \end{array}$$

- (b) We now prove the right-to-left implication:

$$\begin{array}{c}
 \frac{\overline{\forall x.\exists y.(p(x) \vee q(x, y))}}{\exists y.(p(x) \vee q(x, y))} \quad 1 \quad [\forall E] \quad \frac{\overline{p(x) \vee q(x, y)}}{p(x) \vee \exists y.q(x, y)} \quad 2 \quad [\vee I_L] \quad \frac{\overline{p(x)}}{p(x) \vee \exists y.q(x, y)} \quad 3 \quad [\vee I_L] \quad \frac{\overline{q(x, y)}}{\exists y.q(x, y)} \quad 4 \quad [\exists I] \quad \frac{\overline{p(x) \vee \exists y.q(x, y)}}{p(x) \vee \exists y.q(x, y)} \quad [\vee I_R] \quad \frac{\overline{p(x) \vee \exists y.q(x, y)}}{q(x, y) \rightarrow (p(x) \vee \exists y.q(x, y))} \quad 4 \quad [\rightarrow I] \\
 \frac{p(x) \vee \exists y.q(x, y)}{p(x) \vee \exists y.q(x, y)} \quad 2 \quad [\exists E] \\
 \frac{p(x) \vee \exists y.q(x, y)}{\forall x.(p(x) \vee \exists y.q(x, y))} \quad [\forall I] \\
 \frac{\forall x.(p(x) \vee \exists y.q(x, y))}{(\forall x.\exists y.(p(x) \vee q(x, y))) \rightarrow (\forall x.(p(x) \vee \exists y.q(x, y)))} \quad 1 \quad [\rightarrow I]
 \end{array}$$

2. • Left-to-right:

$$\begin{array}{c}
 \frac{\overline{\forall x.\forall y.q(x, y) \rightarrow p(x)}}{\forall y.q(x, y) \rightarrow p(x)} \quad 1 \quad [\forall E] \quad \frac{\overline{\forall y.q(x, y) \rightarrow p(x)}}{q(x, y) \rightarrow p(x)} \quad [\forall E] \quad \frac{\overline{q(x, y)}}{p(x)} \quad 3 \quad [\rightarrow E] \\
 \frac{\exists y.q(x, y)}{p(x)} \quad 2 \quad [\exists E] \\
 \frac{p(x)}{(\exists y.q(x, y)) \rightarrow p(x)} \quad 2 \quad [\rightarrow I] \\
 \frac{(\exists y.q(x, y)) \rightarrow p(x)}{\forall x.(\exists y.q(x, y)) \rightarrow p(x)} \quad [\forall I] \\
 \frac{\forall x.(\exists y.q(x, y)) \rightarrow p(x)}{(\forall x.\forall y.q(x, y) \rightarrow p(x)) \rightarrow \forall x.(\exists y.q(x, y)) \rightarrow p(x)} \quad 1 \quad [\rightarrow I]
 \end{array}$$

- right-to-left:

$$\begin{array}{c}
 \frac{\overline{\forall x.(\exists y.q(x, y)) \rightarrow p(x)}}{(\exists y.q(x, y)) \rightarrow p(x)} \quad 1 \quad [\forall E] \quad \frac{\overline{q(x, y)}}{\exists y.q(x, y)} \quad 2 \quad [\exists I] \\
 \frac{p(x)}{q(x, y) \rightarrow p(x)} \quad 2 \quad [\rightarrow I] \\
 \frac{q(x, y) \rightarrow p(x)}{\forall y.q(x, y) \rightarrow p(x)} \quad [\forall I] \\
 \frac{\forall y.q(x, y) \rightarrow p(x)}{\forall x.\forall y.q(x, y) \rightarrow p(x)} \quad [\forall I] \\
 \frac{\forall x.\forall y.q(x, y) \rightarrow p(x)}{(\forall x.(\exists y.q(x, y)) \rightarrow p(x)) \rightarrow \forall x.\forall y.q(x, y) \rightarrow p(x)} \quad 1 \quad [\rightarrow I]
 \end{array}$$

3. • Left-to-right:

$$\begin{array}{c}
\frac{\frac{\frac{\overline{(\exists x.p(x)) \wedge (\exists y.q(y))}}{\exists x.p(x)} \quad 1 \quad [\wedge E_L] \quad \frac{\frac{\overline{(\exists x.p(x)) \wedge (\exists y.q(y))}}{\exists y.q(y)} \quad 1 \quad [\wedge E_R] \quad \frac{\frac{\overline{p(x)} \quad 2 \quad \overline{q(y)} \quad 3}{p(x) \wedge q(y)} \quad [\wedge I] \quad \frac{\overline{\exists y.p(x) \wedge q(y)}}{\exists x.\exists y.p(x) \wedge q(y)} \quad [\exists I] \quad \frac{\overline{\exists x.\exists y.p(x) \wedge q(y)}}{\exists x.\exists y.p(x) \wedge q(y)} \quad 3 \quad [\exists Ea]}{\exists x.\exists y.p(x) \wedge q(y)} \quad 2 \quad [\exists E]} \quad 1 \quad [\rightarrow I] \\
\frac{\overline{((\exists x.p(x)) \wedge (\exists y.q(y))) \rightarrow \exists x.\exists y.p(x) \wedge q(y)}}{((\exists x.p(x)) \wedge (\exists y.q(y))) \rightarrow \exists x.\exists y.p(x) \wedge q(y)} \quad 1
\end{array}$$

• Right-to-left:

$$\begin{array}{c}
\frac{\frac{\frac{\overline{p(x) \wedge q(y)}}{\exists x.p(x)} \quad 3 \quad [\wedge E_L] \quad \frac{\overline{p(x) \wedge q(y)}}{\exists y.q(y)} \quad 3 \quad [\wedge E_R] \quad \frac{\overline{p(x)} \quad [\exists I] \quad \overline{q(y)} \quad [\exists I]}{\overline{(\exists x.p(x)) \wedge (\exists y.q(y))}} \quad [\wedge I] \quad \frac{\overline{\exists y.p(x) \wedge q(y)} \quad 2 \quad \overline{(\exists x.p(x)) \wedge (\exists y.q(y))}}{\overline{(\exists x.p(x)) \wedge (\exists y.q(y))}} \quad 3 \quad [\exists E]}{\overline{(\exists x.p(x)) \wedge (\exists y.q(y))}} \quad 2 \quad [\exists E]} \quad 1 \quad [\rightarrow I] \\
\frac{\overline{((\exists x.p(x)) \wedge (\exists y.q(y))) \rightarrow ((\exists x.p(x)) \wedge (\exists y.q(y)))}}{((\exists x.p(x)) \wedge (\exists y.q(y))) \rightarrow ((\exists x.p(x)) \wedge (\exists y.q(y)))} \quad 1
\end{array}$$

4. Let P be $(\forall x.p(x) \wedge q(x)) \rightarrow \forall x.p(x)$. To prove that P is valid, we have to prove that for all models M , $\models_M P$ is true. Assume a model M , and let us prove that $\models_M P$ is true. M is of the form $\langle D, \langle \rangle, \langle \mathcal{R}_p, \mathcal{R}_q \rangle \rangle$.

- $\models_M P$
- iff if $\models_M, \forall x.p(x) \wedge q(x)$ then $\models_M, \forall x.p(x)$
- iff if $\models_M, \forall x.p(x) \wedge q(x)$ then for all $d \in D$, $\models_{M, x \mapsto d} p(x)$
- iff if $\models_M, \forall x.p(x) \wedge q(x)$ then for all $d \in D$, $\langle \llbracket x \rrbracket_{x \mapsto d}^M \rangle \in \mathcal{R}_p$
- iff if $\models_M, \forall x.p(x) \wedge q(x)$ then for all $d \in D$, $\langle d \rangle \in \mathcal{R}_p$
- iff if (for all e , $\models_{M, x \mapsto e} p(x) \wedge q(x)$) then for all $d \in D$, $\langle d \rangle \in \mathcal{R}_p$
- iff if (for all e , $\models_{M, x \mapsto e} p(x)$ and $\models_{M, x \mapsto e} q(x)$) then for all $d \in D$, $\langle d \rangle \in \mathcal{R}_p$
- iff if (for all e , $\langle \llbracket x \rrbracket_{x \mapsto e}^M \rangle \in \mathcal{R}_p$ and $\langle \llbracket x \rrbracket_{x \mapsto e}^M \rangle \in \mathcal{R}_q$) then for all $d \in D$, $\langle d \rangle \in \mathcal{R}_p$
- iff if (for all e , $\langle e \rangle \in \mathcal{R}_p$ and $\langle e \rangle \in \mathcal{R}_q$) then for all $d \in D$, $\langle d \rangle \in \mathcal{R}_p$
- iff True
 - Indeed assume (for all e , $\langle e \rangle \in \mathcal{R}_p$ and $\langle e \rangle \in \mathcal{R}_q$) and $d \in D$ and let us prove $\langle d \rangle \in \mathcal{R}_p$
 - Let us instantiate the “for all” assumption with d . We obtain that $\langle d \rangle \in \mathcal{R}_p$ and $\langle d \rangle \in \mathcal{R}_q$. Therefore, our conclusion $\langle d \rangle \in \mathcal{R}_p$ is true.

5. For example, the models

- $M_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n+1 \rangle, \langle \{ \langle n \rangle \mid n \text{ is even} \}, \{ \langle n \rangle \mid n \text{ is odd} \} \rangle \rangle$
- $M'_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n+1 \rangle, \langle \{ \langle n \rangle \mid n \text{ is odd} \}, \{ \langle n \rangle \mid n \text{ is even} \} \rangle \rangle$
- and $M''_1 = \langle \{0\}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \{ \langle n \rangle \mid \text{True} \} \rangle \rangle$

are models of $\exists x.\text{odd}(x) \wedge \neg \text{even}(x)$;

The models

- $M_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n \rangle \mid n \text{ is even} \}, \{ \langle n \rangle \mid n \text{ is even} \} \rangle \rangle$
- $M'_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n \rangle \mid n \text{ is odd} \}, \{ \langle n \rangle \mid n \text{ is odd} \} \rangle \rangle$
- and $M''_2 = \langle \{0\}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n \rangle \mid \text{True} \}, \emptyset \rangle \rangle$

are models of $\neg \exists x. \text{odd}(x) \wedge \neg \text{even}(x)$.